NICK WALKER

Robotics Researcher

nswalker@mit.edu
nickwalker.us

EDUCATION

Ph.D. Computer Science	*University of Washington, Seattle, WA*	2018–25
M.S. Computer Science	*University of Washington, Seattle, WA*	2018–20
B.S.A. Computer Science	*The University of Texas, Austin, TX*	2014–18

APPOINTMENTS

Postdoctoral Research Associate	*Massachusetts Institute of Technology*	2025–

PUBLICATIONS

Thesis

[t1] "Making Robot Behaviors Automatically Transparent." N. Walker. *The University of Washington*. Mar. 2025.

Conference

[c15] "Asking Smarter Questions: Targeted Active Preference Learning for Underspecified Rewards." H. Merker, N. Walker, A. Bobu. *CoRL* 2026.

[c14] "Interpretable Robot Failure Attribution for Remote Robot Supervisors." N. Walker, Z. Wang, M. Grotz, M. Cakmak. *RO-MAN* 2026.

[c13] "Robots That Know What to Ask: Recovering Misaligned Rewards through Targeted Explanations." H. Merker, N. Walker, A. Bobu. *RSS* 2026.

[c12] "I Can Tell What I Am Doing: Toward Real-World Natural Language Grounding of Robot Experiences." Z. Wang, B. Liang, V. Dhat, Z. Brumbaugh, N. Walker, R. Krishna, M. Cakmak. *CoRL* 2024.

[c11] "Fast Explicit-Input Assistance for Teleoperation in Clutter." N. Walker, X. Yang, A. Garg, M. Cakmak, D. Fox, C. Pérez-D'Arpino. *IROS* 2024.

[c10] "Using 3D Mice to Control Robot Manipulators." V. Dhat, N. Walker, M. Cakmak. *HRI* 2024.

[c9] "Not All Who Wander Are Lost: A Localization-Free System for In-the-Wild Mobile Robot Deployments." A. Nanavati*, N. Walker*, L. Taber, C. Mavrogiannis, L. Takayama, M. Cakmak, S. Srinivasa. *HRI* 2022.

[c8] "Influencing Behavioral Attributions to Robot Motion During Task Execution." N. Walker, C. Mavrogiannis, S. Srinivasa, M. Cakmak. *CoRL* 2021.

[c7] "Learning Backchanneling Behaviors for a Social Robot via Data Augmentation from Human-Human Conversations." M. Murray, N. Walker, A. Nanavati, P. Alves-Oliveira, N. Filippov, A. Sauppe, B. Mutlu, M. Cakmak. *CoRL* 2021.

[c6] "Human Perceptions of a Curious Robot that Performs Off-Task Actions." N. Walker, K. Weatherwax, J. Alchin, L. Takayama, M. Cakmak. *HRI* 2020.

[c5] "Open-World Reasoning for Service Robots." Y. Jiang*, N. Walker*, J. Hart, P. Stone. *ICAPS* 2019.

[c4] "Improving Grounded Natural Language Understanding through Human-Robot Dialog." J. Thomason, A. Padmakumar, J. Sinapov, N. Walker, Y. Jiang, H. Yedidsion, J. Hart, P. Stone, R. J. Mooney. *ICRA* 2019.

[c3] "PRISM: Pose Registration for Integrated Semantic Mapping." J. W. Hart, R. Shah, S. Kirmani, N. Walker, K. Baldauf, N. John, P. Stone. *IROS* 2018.

[c2] **"Automatic Curriculum Graph Generation for Reinforcement Learning Agents."** M. Svetlik, M. Leonetti, J. Sinapov, R. Shah, N. Walker, P. Stone. *AAAI* 2017.

[c1] **"Wearable ear EEG for brain interfacing."** E. D. Schroeder, N. Walker, A. S. Danko. *Proc. of SPIE 10051, Neural Imaging Sensing*. Feb. 2017.

Preprint

[a2] **"A Few Words Go a Long Way: Language Guided Robot Policy Synthesis."** D. Chen, A. R. Jain, E. Goossen, E. Romig, M. Murray, N. Walker, M. Cakmak. *arXiv*:2607.23784, Jul. 2026.

[a1] **"An Architecture for Person-Following using Active Target Search."** M. Kim, M. Arduengo, N. Walker, Y. Jiang, J. W. Hart, P. Stone, L. Sentis. *arXiv*:1809.08793, Sept. 2019.

Journal

[j1] **"Jointly Improving Parsing and Perception for Natural Language Commands through Human-Robot Dialog."** J. Thomason, A. Padmakumar, J. Sinapov, N. Walker, Y. Jiang, H. Yedidsion, J. Hart, P. Stone, R. J. Mooney. *JAIR* 2020.

Patent

[p1] **"Transcription analysis platform."** M. J. Szentes, C. Chavez, R. E. Lewis, N. S. Walker. *US11837214*, Dec. 2023.

Refereed Symposium, Workshop

[w7] **"Towards Personalizing Shared Autonomy with Human-in-the-Loop Robot Programs."** D. Chen, S. Vuddagiri, C. Yang, N. Walker, M. Cakmak. *CoRL H2R* 2025.

[w6] **"Can Large Language Models Help Developers with Robotic Finite State Machine Modification?."** X. Gan*, Y. R. Song*, N. Walker, M. Cakmak. *CoRL* 2024.

[w5] **"Towards robustly picking unseen objects from densely packed shelves."** M. Grotz, J. Lowry, S. Atar, Y. Li, P. Torrado, B. Yang, N. Walker, M. Murray, D. Fox, M. Cakmak, J. R. Smith. *Proc. RSS Workshop Perception Manipulation Challenges for Warehouse Automation*. Jul. 2023.

[w4] **"Desiderata for Planning Systems in General-Purpose Service Robots."** N. Walker*, Y. Jiang*, M. Cakmak, P. Stone. *Proc. of 2019 ICAPS Workshop Planning Robotics*. Jul. 2019.

[w3] **"Neural Semantic Parsing with Anonymization for Command Understanding in General-Purpose Service Robots."** N. Walker, Y.-T. Peng, M. Cakmak. *RoboCup 2019: Robot Soccer World Cup XXIII*. Jul. 2019.

[w2] **"LAAIR: A Layered Architecture for Autonomous Interactive Robots."** Y. Jiang*, N. Walker*, M. Kim, N. Brissonneau, D. S. Brown, J. W. Hart, S. Niekum, L. Sentis, P. Stone. *AAAI Fall Symp. Reasoning Learning in Real-World Systems for Long-Term Autonomy*. Oct. 2018.

[w1] **"Interaction and Autonomy in RoboCup@Home and Building-Wide Intelligence."** J. Hart, H. Yedidsion, Y. Jiang, N. Walker, R. Shah, J. Thomason, A. Padmakumar, R. Fernandez, J. Sinapov, R. Mooney, P. Stone. *AAAI Fall Symp. Artificial Intelligence Human-Robot Interaction*. Oct. 2018.

Periodical Feature

[f2] **"A Guide to Transit-Oriented Running in Seattle."** N. Walker. *The Urbanist*, Nov. 2023.

[f1] **"Wandering Robots in the Wild."** N. Walker, A. Nanavati. *IEEE Spectrum*, Jul. 2022.

RESEARCH EXPERIENCE

Postdoctoral Research Associate *Massachusetts Institute of Technology* 2025–
- Improving methods for reinforcement and imitation learners to communicate their uncertainty [c13, c15]

Graduate Research Assistant *University of Washington* 2018–25
- Learning models that help robot supervisors diagnose failures [c14]
- Generating communicative motion and language during task execution [c8, c12, t1]
- Perceptions of intrinsically motivated robot behaviors [c6, c8]

Research Intern *NVIDIA* 2022 Su.
- Designed and evaluated teleoperation assistance for manipulation in clutter [c11]

Research Engineer Intern *USAA* 2016 Su.
- Ear-worn brain-computer interface software and hardware for biometric authentication [c1]

Undergraduate Research Assistant *University of Texas* 2016–18
- Long-term autonomy for service robots [c5, w1, w2, w4]
- Mobile manipulation in homes and offices [j1, a1, w1]
- Grounded natural language understanding [c4, j1, w3]
- Automated curriculum learning for reinforcement learning agents [c2]

Research Engineer Intern *USAA* 2015 Su.
- Evaluation of automated speech transcription vendors [p1]

GRANTS

Documenting Geyser Basin on the University of Washington Campus 2025
4Culture, Preservation Special Projects Grant. $9,500. PI.

A Speech and Language Dataset of GPSR Commands 2020
RoboCup Federation, League Development Grant. $2,200. PI.

TEACHING EXPERIENCE

Teaching Assistant *UW CSE 478 (Robotics)* 2021 Sp.
- Updated assignments, developed unit tests and CI-based autograder used in 5+ offerings

Teaching Assistant *UW CSE 481C (Robotics Capstone)* 2019 Wi.
- Developed assignments, supported undergraduates using the Kuri robot

MENTORING

Michelle Wang	Undergraduate Researcher	2026–
Jocelyn Zhao	Undergraduate Researcher	2026–
Daphne Chen	Graduate Researcher	2025–
Helena Merker	Graduate Researcher	2025–26
Zihan Wang	Graduate Researcher » Meta	2023–25
Varad Dhat	Graduate Researcher » Torc Robotics	2023–25
Joe Sluis	Undergraduate Researcher » startup	2022–25
Brian Yao	Undergraduate Researcher » Amazon	2022–25
Xiangyu Gan	Undergraduate Researcher » startup	2023–24
Yuxin Ray Song	Undergraduate Researcher » Researcher, MIT	2023–24
Zachary Brumbaugh	Undergraduate Researcher » startup	2022–23

Kavi Dey	Highschool Intern » Harvey Mudd	2021–22
Aidan Alchin	Highschool Intern » Stanford	2019

LEADERSHIP AND PROFESSIONAL SERVICE

Director	Drumheller Marathon & Half Marathon	2022–
Director	Light Rail Relay	2021–
Organizer	Northwest Robotics Symposium	2022
Organizer	NSF GRFP Seminar, Allen School, UW	2020
Organizer	Practical Service Robots Workshop, RSS	2020
Organizer	Imitation Learning Workshop, RSS	2020
Technical Committee	RoboCup@Home	2019–20
Peer Mentor	Allen School First Year Graduate Student Mentoring, UW	2019–21
Reader	Allen School Ph.D. Admissions Committee, UW	2018

Reviewing

HRI	’27–’22	RO-MAN	’26	IJSR	’25, ’23–’22
ICRA	’27–’23, ’21, ’19	RSS	’26, ’21	IROS	’24, ’21
CoRL	’26–’22	THRI	’26–’23, ’20	T-RO	’24
RA-L	’26–’25, ’21	CHI	’25	Sci. Rob.	’21

RECOGNITION

Best Paper: Short Contributions — *ACM/IEEE Intl. Conf. Human-Robot Interaction* — 2024

Graduate Research Fellowship — *National Science Foundation* — 2020–25

Computer Science & Engineering Research Fellowship — 2018–19
Allen School, UW

Best Poster, with UT Austin Villa — *RoboCup@Home DSPL* — 2018

Commencement Student Speaker — *College of Natural Sciences, UT* — 2018

GRFP Honorable Mention — *National Science Foundation* — 2018

Dean’s Honored Graduate — *College of Natural Sciences, UT* — 2018

Outstanding Undergraduate Researcher Award Honorable Mention — 2018
Computing Research Association

TIDES Fellowship — *Texas Institute for Discovery Education in Science, UT* — 2017

College of Natural Sciences Scholarship — 2014–18
College of Natural Sciences, UT

SKILLS

Languages	Python, C++, Javascript/HTML/CSS, Answer Set Programming
Frameworks & Tools	PyTorch, ROS, Docker, Isaac Sim
Digital media	Premiere, Photoshop, Illustrator, InDesign, Lightroom

PERSONAL

nickwalker.us
github.com/nickswalker
orcid.org/0000-0001-7711-0003